

Spencer Hum

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EDUCATION

Queen's University

Sep. 2022 – Present

Bachelor of Computing (Honours) with Professional Internship

Kingston, ON

- Queen's University Excellence Scholarship, Dean's Honour List

WORK EXPERIENCE

Yupi.ca

May 2025 – Present

Software Engineer Intern

Kingston, ON

- Drove continuous improvement on the internal ERP system by implementing end-to-end workflows for ingesting contracts, updating inventory, and logging sales using React and a Python/Django backend
- Designed and delivered a full-stack time-tracking application to enable seamless work-log entry for employees and real-time oversight for management to monitor project progress
- Implemented a company-wide tracking system for employees to log hours and business expenses, standardizing data entry across 3 locations and significantly reducing payroll processing time using Django
- Cut manual search time for supplier data from 2 hours to 30 minutes per day by creating a Django-based query interface that filters supplier and trade data using custom SQL queries
- Developed a centralized permissions-assignment service using Role-Based Access Control (RBAC) to secure sensitive company data within the core ERP platform, automating access for 100+ employees
- Automated report generation using ETL pipelines to create SQL-based reports to track inventory and sales
- Conducted user interviews to identify operational pain-points, delivering targeted UI/UX enhancements and backend optimizations such as filter options and performance issues

Canada Computers & Electronics

Sept. 2024 – Dec. 2024

Software Engineer Intern – B2B Platform

Richmond Hill, ON

- Developed and optimized the B2B e-commerce storefront using Node.js and TypeScript, deploying REST APIs that efficiently handle 1000+ concurrent requests across enterprise clients
- Integrated distributor APIs allowing corporate clients to browse and purchase inventory directly through suppliers
- Designed access control for different roles such as admin, accounting and purchasing for organizations using Express.js, enabling organizations to self-manage users and reducing support tickets from over 50 to 25 per month
- Automated computation for corporate discounts and bulk rates based on distributor cost data using TypeScript and MySQL, ensuring consistent quote accuracy across 10K+ SKUs

PROJECTS

LLM Compare | *Nuxt, Vue.js, TypeScript, Hono, Node.js*

- Built a full-stack LLM benchmarking platform that compares the responses between different LLMs under prompts in real-time using TypeScript and Hono with Vercel's AI SDK
- Implemented shared Zod contracts with Hono to validate all LLM requests and responses, guaranteeing consistent API behavior across non-deterministic model outputs
- Maximized type safety by eliminating type assertions to catch errors at compile time instead of runtime

Agentic RAG Engine | *Python, LangGraph, FastAPI, PostgreSQL, Docker*

- Developed an autonomous Agentic RAG using LangGraph and the OpenAI API with a self-correcting feedback loop that evaluates and improves model responses to reduce AI hallucinations
- Built a scalable production-grade data pipeline with Docker and PostgreSQL (pgvector) to ingest, chunk, and generate vector embeddings for proprietary enterprise datasets
- Implemented a FastAPI service leveraging asynchronous Python to deliver real-time streaming responses

SKILLS

Languages: Python, Java, JavaScript, TypeScript, SQL, Go, C/C++, HTML, CSS

Technologies: React, Vue.js, Hono, Express.js, Django, Flask, FastAPI, Next.js, Node.js, Terraform, MySQL, PostgreSQL, MongoDB, Kafka, Redis, AWS (Amazon Web Services), Docker

Certifications: AWS Certified Cloud Practitioner